LECTURE 3
HYBRIDIZATION OF DIFFERENT ENERGY STORAGE DEVICES.

Because of the frequent stop-and-go operation of EVs and HEVs, the discharging and charging profile of the energy storage is highly varied. The average power required from the energy storage is much lower than the peak power for acceleration and hill climbing in a relatively short duration. The ratio of peak power to average power can reach over 10:1. In HEV design, the peak power capacity of the energy storage is more important than its energy capacity, and usually constrains its size reduction. Based on present battery technology, battery design has to carry out the trade-off among specific energy, specific power, and cycle life. The difficulty in simultaneously obtaining high values of specific energy, specific power, and cycle life has led to some suggestions that the energy storage system of EV and HEV should be a hybridization of an energy source and a power source.

Hybridization of energy storage devices is the process of combining two or more energy sources or storage devices to meet the energy and power requirements of an electric vehicle (EV). Different energy-storage devices have different characteristics. Batteries have relatively high specific energy but may have limitations in supplying sudden high-power demands. Ultracapacitors have high specific power and can respond quickly to changes in power demand, but they store less energy. Flywheels store energy mechanically and can deliver or absorb power rapidly, while fuel cells generate electricity continuously as long as fuel is supplied.

The energy source, mainly batteries and fuel cells, has high specific energy, whereas the power source has high specific power. Power sources can be recharged from the energy source during less demanding driving or regenerative braking. The power source that has received wide attention is the ultracapacitor

Ragone Plot and Its Importance in Electric Vehicles

A Ragone plot is a graphical representation used to compare the performance of different energy-storage and energy-conversion sources based on their specific energy and specific power. It is particularly useful in EV applications because an EV energy source must satisfy two important requirements: it should store sufficient energy to provide a long driving range, and it should be capable of delivering sufficient power for acceleration and other high-power demands.

The Ragone plot shown is a logarithmic plot, which allows energy sources having very different ranges of specific energy and specific power to be represented on the same graph. In the figure, specific power is plotted on the horizontal axis in W/kg, while specific energy is plotted on the vertical axis in Wh/kg. The logarithmic scale is necessary because the values extend over several orders of magnitude. The diagonal lines represent approximately constant discharge time, since

$$t = \frac{SE}{SP}$$

where SE is the specific energy and SP is the specific power. Thus, the position of an energy source on the Ragone plot provides information about both the amount of energy it can store and the rate at which that energy can be delivered.

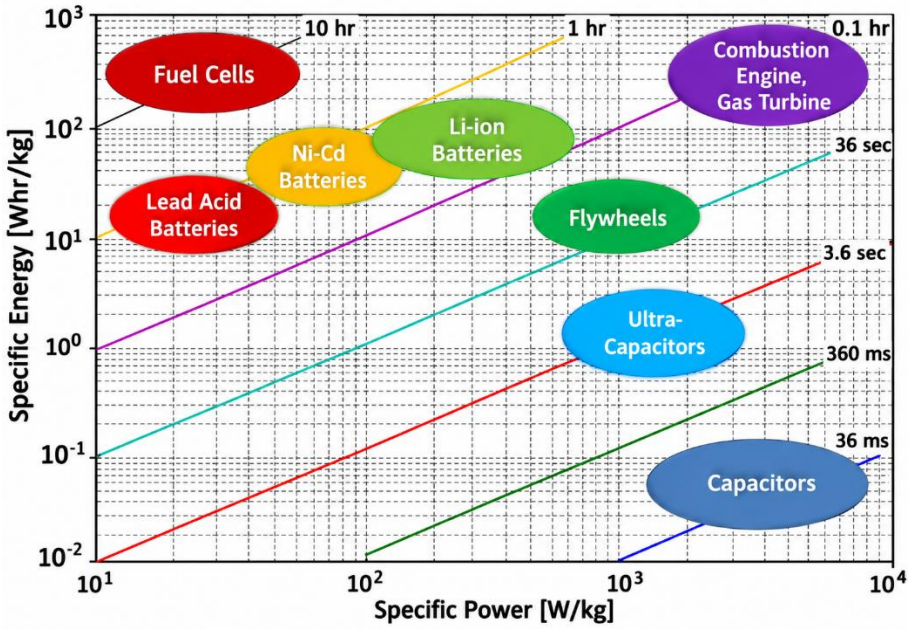


Figure 1: Ragone Plot

From the plot, it can be observed that different energy sources have different combinations of specific energy and specific power. Fuel cells and batteries generally provide relatively high specific energy, making them suitable for applications requiring sustained energy delivery. Among the battery technologies shown, Li-ion batteries occupy a region with relatively higher specific energy and power compared with lead-acid and Ni-Cd batteries. On the other hand, flywheels and ultracapacitors can provide relatively high specific power, making them useful when rapid energy delivery or absorption is required. Conventional capacitors can provide very high power over extremely short durations, but their specific energy is comparatively low.

An important observation from the Ragone plot is that no single energy source simultaneously provides the ideal combination of very high specific energy and very high specific power. An energy source designed primarily for high energy storage may not be able to supply very high power for rapid acceleration, while a source capable of delivering very high power may store comparatively little energy. This difference creates an important design consideration for electric vehicles. Thus, the Ragone plot is an important tool for EV energy-source selection and system design. It provides a visual comparison of specific energy and specific power and helps engineers understand the trade-off between driving range and high-power performance. The plot also provides the fundamental motivation for combining batteries with high-power devices such as ultracapacitors or flywheels in hybrid energy-storage systems.

For this reason, hybrid energy-storage systems are increasingly considered for EV applications. In such a system, two or more energy sources are combined so that their characteristics complement each other. For example, a battery can act as the main energy source, providing the energy required for driving range, while an ultracapacitor or flywheel can support short-duration high-power demands such as acceleration and regenerative braking. The hybrid system can therefore provide both adequate energy and high power while reducing the need for a single source to satisfy all operating requirements.

CONCEPT OF HYBRID ENERGY STORAGE

The hybridization of energy storage involves to combining two or more energy storages together so that the advantages of each can be brought out and the disadvantages can be compensated by others. For instance, the hybridization of a chemical battery with an ultracapacitor can overcome problems such as the low specific power of chemical batteries and low specific energy of ultracapacitors, thus achieving high specific energy and high specific power. Basically, the hybridized energy storage consists of two basic energy storages, one with high specific energy and the other with high specific power. The basic operation of this system is illustrated in Figure 2. In high power demand operation, such as acceleration and hill climbing, both basic energy storages deliver their power to the load as shown in Figure 2a. On the other hand, in low-power demand operation, such as constant-speed cruising operation, the high specific energy storage will deliver its power to the load and charge the high specific power storage to recover its charge lost during high-power demand operation, as shown in Figure 2b. In regenerative braking operation, the peak power will be absorbed by the high specific power storage, and only a limited part is absorbed by the high specific energy storage. In this way, the whole system would be much smaller in weight and size than if any one of them alone was the energy storage.

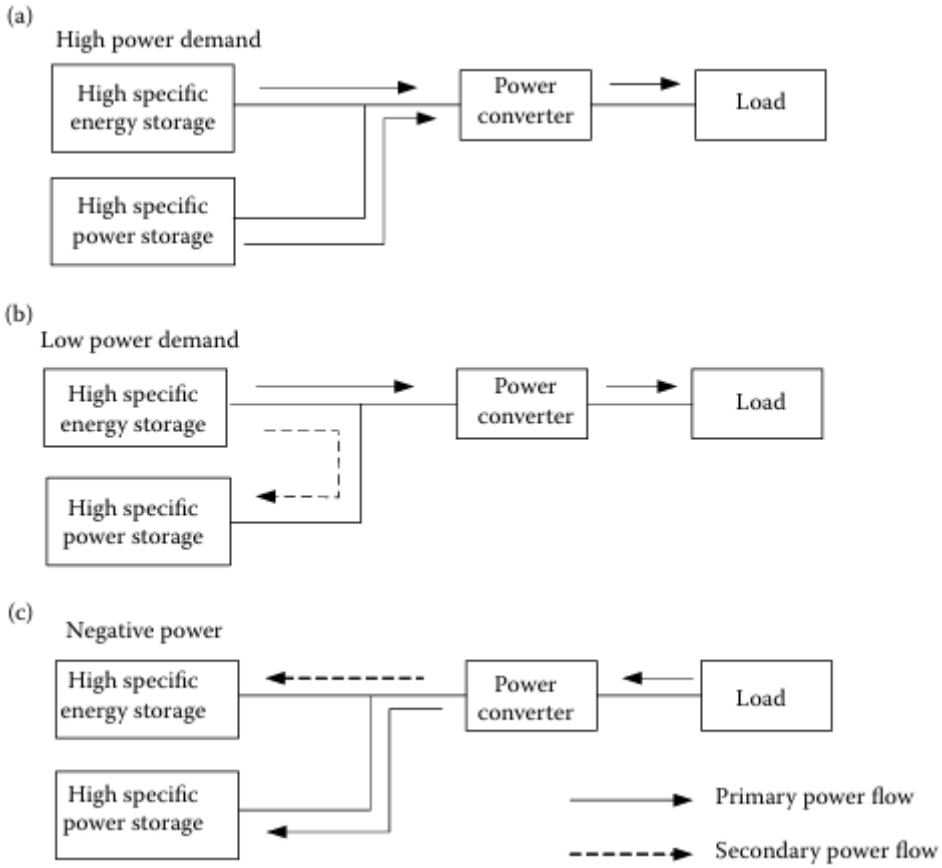


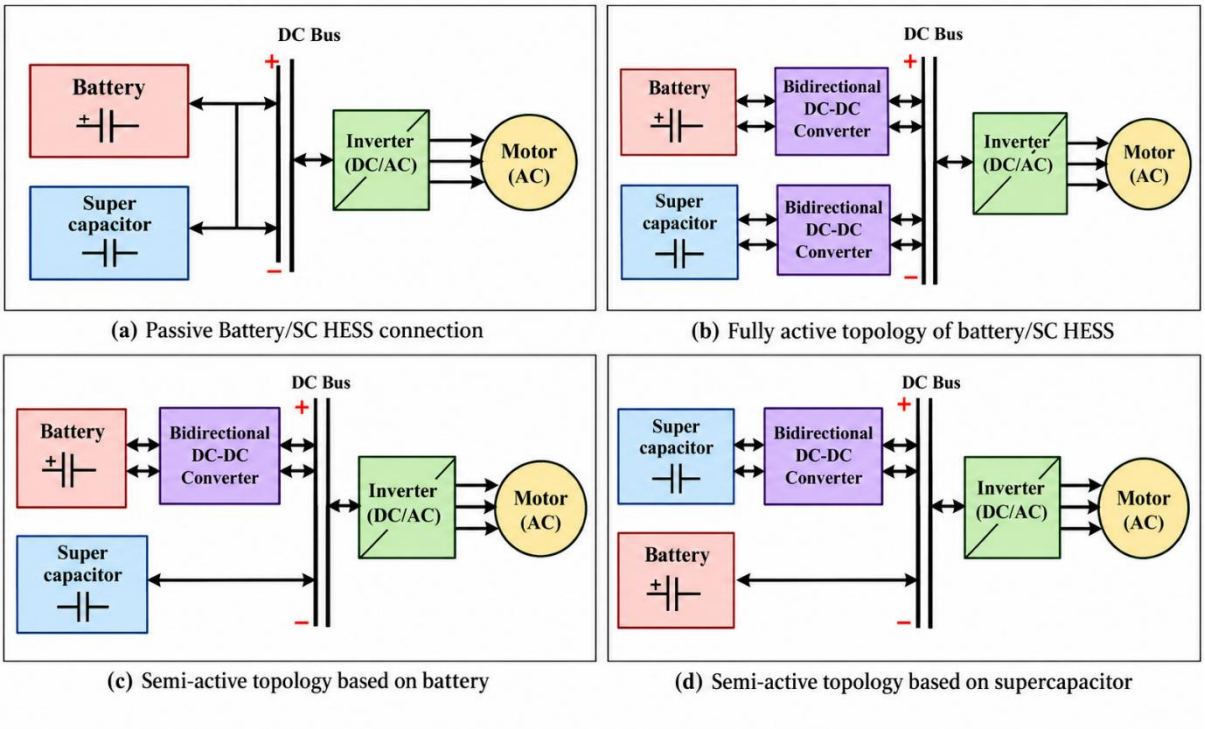
Figure 2: Concept of a hybrid

HYBRIDIZATION OF BATTERY AND ULTRACAPACITOR

A battery–ultracapacitor hybrid system is a common energy-storage arrangement for EVs and HEVs. The battery provides the main energy required for driving, while the ultracapacitor supplies or absorbs high power during short periods. Since ultracapacitors have high specific power but relatively low specific energy, they complement the energy-storage characteristics of batteries.

During acceleration, the ultracapacitor can provide additional power and reduce peak current demand on the battery. During regenerative braking, it can absorb a large amount of power in a short time. This arrangement can reduce battery current fluctuations and voltage drops.

There are two main ways to connect a battery and an ultracapacitor: passive hybridization and active hybridization.



(a) Passive Battery/SC HESS connection

(b) Fully active topology of battery/SC HESS

(c) Semi-active topology based on battery

(d) Semi-active topology based on supercapacitor

Passive Hybrid Energy Storage System

In a passive hybrid system, the battery and ultracapacitor are connected directly in parallel without a DC–DC converter between them. Both devices share the same terminal voltage, and the power distribution between them depends mainly on their electrical characteristics and operating conditions.

The ultracapacitor acts as a current filter. It helps reduce peak battery current and can reduce the battery's voltage drop during sudden increases in power demand. During regenerative braking, it can also accept some of the recovered energy.

The main advantage of passive hybridization is its simple construction and low cost because it does not require a separate converter to control power flow between the battery and ultracapacitor. However, the power flow cannot be controlled independently, and the energy stored in the ultracapacitor may not be fully utilized because its voltage is constrained by the battery voltage.

Active Hybrid Energy Storage System

In an active hybrid system, a bidirectional DC–DC converter is placed between the battery and ultracapacitor. The converter allows the two devices to operate at different voltages and controls the power transferred between them.

During acceleration, the controller can regulate the ultracapacitor's contribution to meet peak power demand. During regenerative braking, the converter controls the flow of recovered energy into the ultracapacitor. It can also regulate the transfer of energy between the two storage devices according to their operating conditions.

A two-quadrant or bidirectional converter allows power to flow in both directions. This improves control over charging and discharging and allows more effective use of the ultracapacitor's stored energy.

The main advantage of active hybridization is better power management and improved utilization of the energy-storage devices. Its disadvantages are higher cost, increased circuit complexity, and the need for an appropriate control strategy.

Parameter	Passive system	Active system
Connection	Direct parallel connection	Connected through a DC–DC converter
Voltage	Both devices share terminal voltage	Devices can operate at different voltages
Power-flow control	Limited; no independent converter control	Active control is possible
Ultracapacitor energy utilization	May be incomplete	Can be improved
Cost and complexity	Lower	Higher
Control requirements	Simple	More sophisticated

HYBRID FUEL CELL ENERGY STORAGE SYSTEMS

A Proton Exchange Membrane Fuel Cell (PEMFC) can provide continuous electrical power, but its response to sudden changes in power demand may be too slow for vehicle applications. Its output voltage also requires regulation. Designing a fuel cell large enough to supply every peak power demand can increase system cost and size.

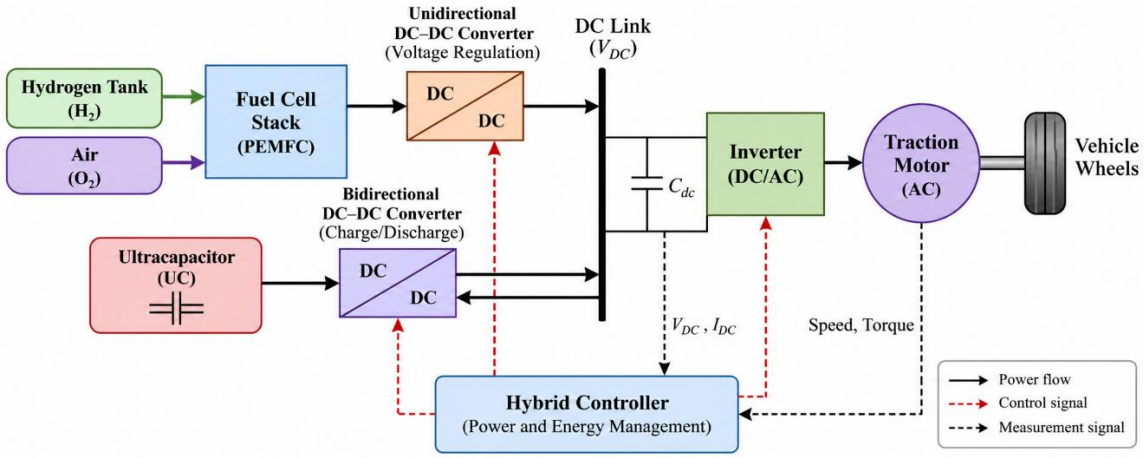
To overcome these limitations, fuel cells are combined with batteries or ultracapacitors. Fuel cells have high specific energy when the fuel storage system is considered, while ultracapacitors have high specific power. Batteries provide an intermediate combination of energy and power characteristics. A fuel cell–ultracapacitor system can provide continuous energy from the fuel cell and short-duration peak power from the ultracapacitor. A fuel cell–battery–ultracapacitor system can offer additional flexibility because the battery stores more energy than an ultracapacitor, while the ultracapacitor handles rapid power fluctuations. The battery can also provide the electrical power required to start up the fuel-cell system.

In a fuel cell–battery hybrid system, the fuel cell supplies relatively steady power, while the battery provides additional power during acceleration and other transient conditions. The battery can also store energy recovered during regenerative braking.The fuel cell and battery can be controlled independently to meet the load requirements. For example, the battery can

regulate the DC-link voltage, while the fuel cell is controlled to supply the required current. This allows the fuel cell to operate under more suitable conditions and reduces the need to oversize it for peak power demand.

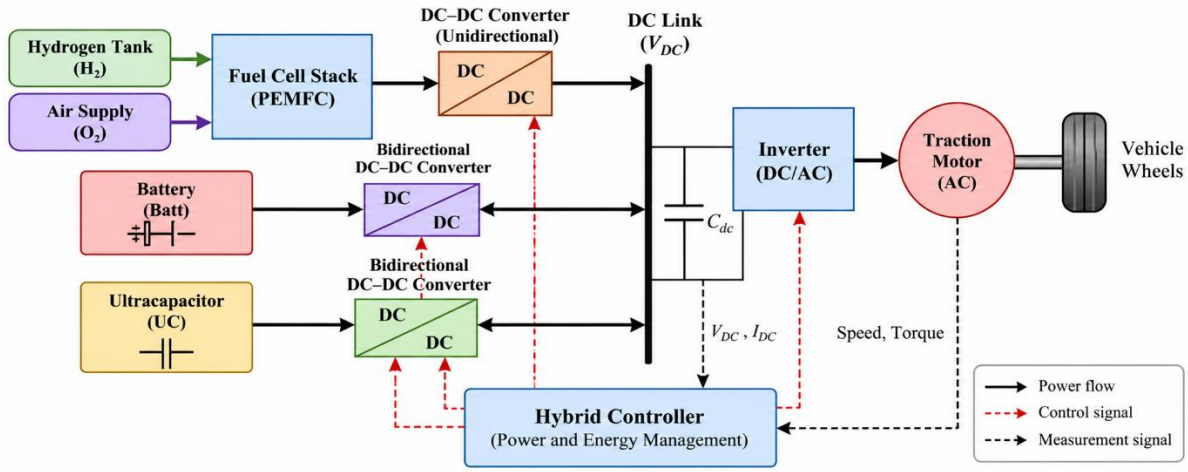
Fuel Cell–Ultracapacitor Hybrid System

In a fuel cell–ultracapacitor hybrid system, the fuel cell supplies the average power required by the vehicle, while the ultracapacitor provides short-duration peak power. During acceleration, the ultracapacitor supplies additional power. During regenerative braking, it absorbs energy rapidly. This arrangement helps reduce sudden power fluctuations experienced by the fuel cell. However, because ultracapacitors store less energy than batteries, their ability to supply power over longer periods is limited.



Fuel Cell–Battery–Ultracapacitor Hybrid System

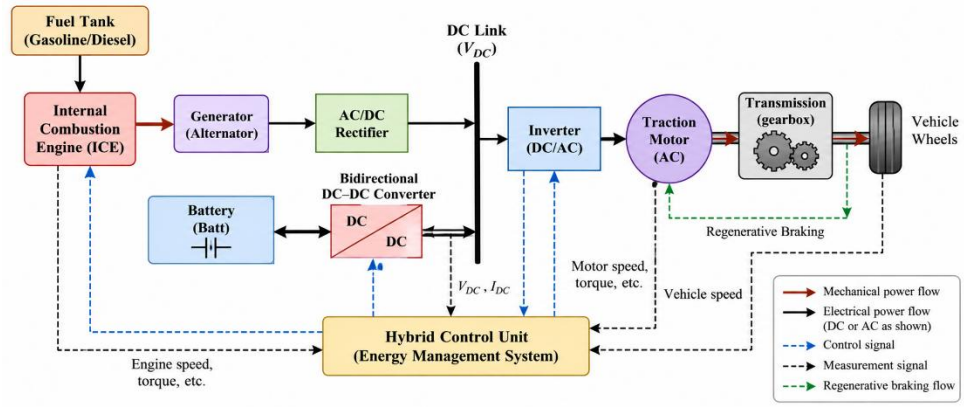
This system combines three devices with complementary characteristics. The fuel cell provides continuous power, the battery stores electrical energy and supplies additional power, and the ultracapacitor handles rapid changes in power demand. During steady driving, the fuel cell supplies the required average power. During acceleration, the battery and ultracapacitor can support the fuel cell. During regenerative braking, the ultracapacitor can absorb high-power pulses, while the battery can store recovered energy according to its charging limits and available capacity. This arrangement offers flexibility in system sizing and power management, but requires a more sophisticated controller.



HYBRIDIZATION OF BATTERIES WITH OTHER ENERGY SOURCES

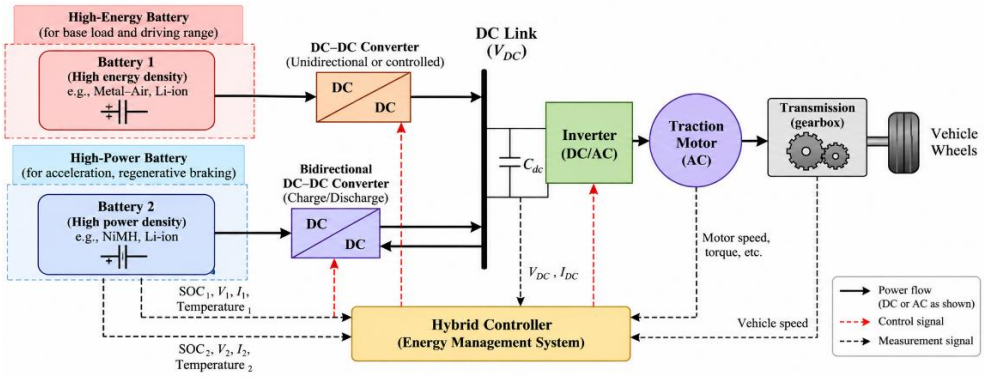
Internal Combustion Engine–Battery Hybrid

An internal combustion engine (ICE)–battery hybrid combines a conventional engine with a rechargeable battery and an electric drive system. The battery can store electrical energy and supply power when required, while the engine provides mechanical power and may also drive a generator to charge the battery, depending on the vehicle configuration. A battery that can be charged and discharged efficiently is desirable for this type of hybrid. Nickel-metal hydride (NiMH) batteries have been used in vehicles such as the Toyota Prius and Honda Insight. The supplied material gives a typical battery capacity of approximately 2–5 kWh for this class of hybrid vehicle. This arrangement allows the engine and battery to work together to improve energy utilization under changing driving conditions.



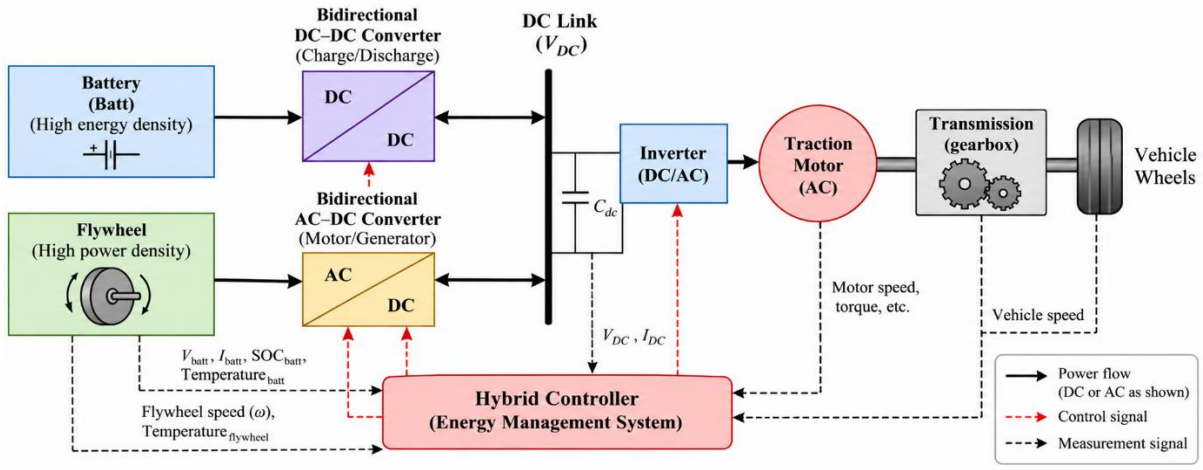
Battery–Battery Hybrid

A battery–battery hybrid combines two different battery technologies to use their complementary characteristics. Different batteries have different specific energy, specific power, charging capability, and discharge characteristics. For example, an aluminium–air battery has high energy-storage potential but limited power capability and cannot be electrically recharged in the vehicle. It can therefore be combined with a rechargeable battery, such as a nickel-metal hydride battery. In this arrangement, the aluminium–air battery supplies the base energy demand, while the NiMH battery supplies additional power during acceleration and overtaking. The rechargeable battery can also accept and supply energy during regenerative braking. This configuration combines the energy-storage capability of one battery with the power-delivery and recharging capability of another.



Battery–Flywheel Hybrid

A battery–flywheel hybrid combines electrochemical energy storage with mechanical energy storage. The battery supplies energy for normal driving, while the flywheel stores and releases energy rapidly through a rotating mass. During acceleration, the flywheel can release stored kinetic energy to assist the vehicle. During braking, it can absorb energy by increasing its rotational speed. The stored energy can then be used later to support propulsion. This combination is particularly useful when the battery alone cannot efficiently handle frequent, rapid charging and discharging. For example, a zinc–air battery cannot be electrically recharged onboard in the way a conventional rechargeable battery can, but a flywheel can provide a means of recovering braking energy.



Hybrid configuration	Main energy source	Supporting device	Main purpose
Battery–ultracapacitor	Battery	Ultracapacitor	Peak-power support and regenerative braking
Battery–flywheel	Battery	Flywheel	Rapid mechanical energy recovery
ICE–battery	Engine and battery	Rechargeable battery	Combine engine operation with electric assistance
Battery–battery	High-energy battery	High-power rechargeable battery	Combine energy capacity with rapid power delivery
Fuel cell–battery	Fuel cell	Battery	Continuous power and transient support
Fuel cell–ultracapacitor	Fuel cell	Ultracapacitor	Continuous power and rapid power response
Fuel cell–battery–ultracapacitor	Fuel cell	Battery and ultracapacitor	Combine continuous energy, energy storage, and peak-power capability